At the infrastructure level, the application flow breaks down into:

Frontend (React): A role-aware single-page application (SPA) that routes user actions to appropriate backend endpoints.

API gateway: A routing layer that forwards requests to relevant microservices while enforcing centralised access policies.

Microservices:

- *Auth service:* Handles login, JWT generation, and role-based access.
- *Appointment service:* Manages booking, scheduling, and session updates.
- *Group management:* Supports patient–doctor interaction in structured support groups.
- *Resource service:* Allows content upload and sharing.

Database layer: Each service connects to a corresponding relational database (PostgreSQL), with persistence handled individually for modularity.

Monitoring stack: Logs from every container are shipped to Logstash, indexed via Elasticsearch, and visualised using Kibana (ELK Stack).

Secrets and security: Sensitive configurations like tokens and credentials are managed using Kubernetes secrets and *.env* abstraction during builds.

Figure 1 illustrates the modular design of the application stack, showing how frontend, backend services, CI/CD, and monitoring components interact across the Kubernetes cluster.

This setup not only helped us split development across teams but also gave us a resilient production architecture that is easy to scale, troubleshoot, and monitor. By keeping infrastructure modular and observable, we've been able to test new features, rollback deployments, and handle spikes in user activity without downtime.

CI/CD pipeline with Jenkins, Docker, and Ansible

HealSphere's entire application lifecycle from code commit to production deployment is automated using a combination of GitHub, Jenkins, Docker, Ansible, and Kubernetes. A GitHub webhook notifies Jenkins on every commit to the main branch, automatically triggering the pipeline.

The pipeline is defined in a Jenkinsfile, and runs on a Jenkins agent configured with Docker, kubectl, Ansible, and the Kubernetes Ansible collection. You can find the file at *https://github.com/Kanan-30/HealSphere_Project/blob/main/Jenkinsfile.*

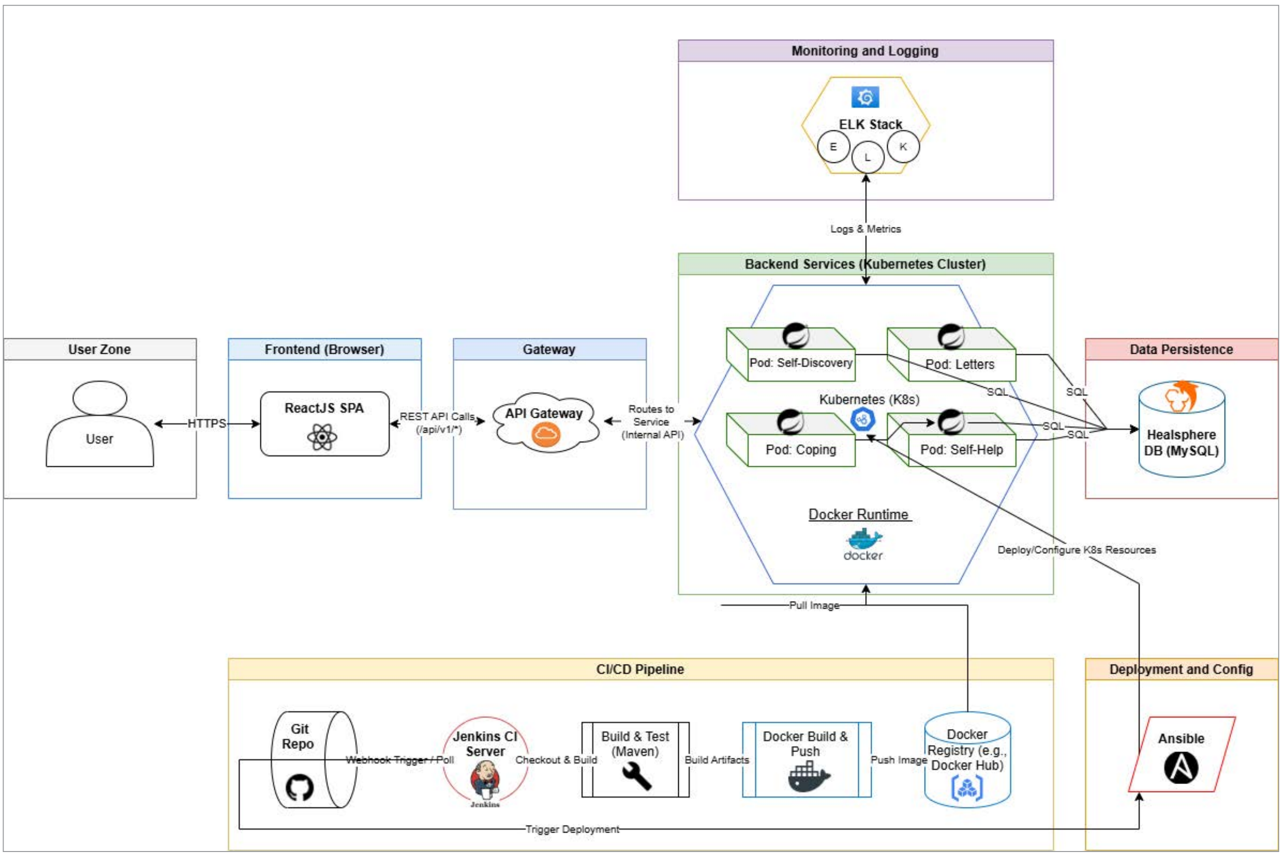

Figure 1: HealSphere system architecture